

Mingye JIANG

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https://jmy2003.github.io/ | https://github.com/JMY2003 | Current Postgraduate Student |
LLM Agent & LLM Algorithms



EDUCATION

2025/09 - 2027/06 Sun Yat-sen University Master - Applied Statistics

School: School of Mathematics
Major: Applied Statistics
Academic Focus: Deep Learning, Large Language Models Relevant
Courses: Selected Topics in Artificial Intelligence Algorithms; Machine Learning and Data Mining; Advanced Programming for Data Analysis; Statistical Inference; Regression Analysis; Algorithm Design...

2021/09 - 2025/06 South China University of Technology Bachelor - Mathematics and Applied Mathematics & Automation

School: School of Mathematics
Major: Mathematics and Applied Mathematics (major); Automation (minor)
Thesis Focus: Distributed Optimization
Courses: Mathematical Analysis; Advanced Algebra; Probability Theory; Mathematical Statistics; C++ Programming; Data Structures and Algorithms; Analog Electronics; Digital Electronics; Mathematical Modeling; Principles of Automatic Control; Artificial Intelligence; Microcomputer Systems and Applications...

PROJECTS

2026/05 - Now Golden Template Intelligent Network Configuration Validation and O&M System Technical Focus: LLM Agent

Golden Template is a Huawei platform built for intelligent validation and operations of network configurations. It aims to improve configuration inspection efficiency and O&M quality through rule-based and automated methods. I designed and built Nora, an intelligent assistant (Network Orchestration & Rules Agent), for the system. Based on natural language understanding, Nora automatically parses users' O&M requests into structured rules, supports rule generation, modification, saving, management, and invocation, and can coordinate with existing system resources to complete configuration validation tasks. The assistant significantly reduces the complexity for network engineers in writing validation rules, running configuration checks, and maintaining rules, improving platform usability and automation.

2026/06 - Now LumenIndex: PageIndex-Inspired Agentic-RAG Multi-Document Retrieval QA System Technical Focus: LLM Agent

Independently designed and open-sourced a self-hostable multi-document retrieval and question-answering system inspired by the open-source PageIndex idea, supporting PDF / DOCX / Markdown multi-document retrieval, persistent chat sessions, and Docker Compose deployment. At the indexing layer, it avoids the expensive path of continuously asking LLMs to build TOCs for large documents, and instead uses deterministic strategies such as PDF bookmarks, layout headings, TOC-page detection, page-level chunking, and large-node refinement. LLM structured outputs are validated with Pydantic, while the agent runtime introduces ReAct-based deep retrieval, automatic context compaction, offline fallback, recoverable errors, and visible indexing progress, turning vectorless reasoning-based RAG into a deliverable system rather than a demo. GitHub: <https://github.com/JMY2003/pageindex-lumenindex-agentic-rag>

2026/03 - 2026/05 Network Configuration Translation and Generation System Based on Agentic RAG Technical Focus: LLM Agent

As a core team member, I built a high-accuracy Agentic RAG tool for configuration translation and generation scenarios, supporting retrieval and generation over Huawei and Cisco device command manuals. In tests, the retrieval accuracy exceeded 98%, much higher than traditional embedding-based retrieval (below 90% and unable to handle similar commands). The system has been deployed in projects including Cisco-to-Huawei configuration file translation, routing network configuration generation, fault injection, and network world models.

2026/03 - 2026/04 Multi-Agent Competitor Comparison and Analysis System Technical Focus: LLM Agent

I designed and implemented a multi-agent competitor analysis platform for general market intelligence research. After users input a product category and analysis objective, the platform automatically completes task understanding, competitor discovery, source planning, web search and crawling, structured information extraction, evidence-chain construction, multi-dimensional competitor analysis, quality review, and report generation. It supports DAG visualization, an evidence repository, report editing, and export, enabling an automated competitor analysis and comparison workflow that is traceable, reviewable, and exportable.

2024/10-2025/6 Research on Distributed Optimization Under Event-Triggered Communication Technical Focus: Optimization Theory

This project focused on distributed gradient tracking methods. For the DIGing algorithm under undirected connected networks, I proposed an improved event-triggered mechanism and derived a new event-triggered communication-based DIGing algorithm. Under strongly convex objective functions, the algorithm converges to the global optimum, theoretically avoiding the parameter selection difficulties of existing triggering mechanisms, making parameter selection simpler and reducing communication burden to some extent. For directed strongly connected networks, I extended this event-triggered mechanism to the Push-DIGing algorithm and designed a new event-triggered Push-DIGing algorithm. Under strongly convex objective functions, the algorithm converges to the global optimum and significantly reduces the communication rate compared with the original Push-DIGing algorithm. This research received the Outstanding Undergraduate Thesis Award of South China University of Technology.

WORK EXPERIENCE

2026/03 - Now Huawei Technologies Co., Ltd. Position: AI Engineer (LLM Agent Focus)

Organization: ICT Business Group (ICT BG), Network Digital Twin Simulation Laboratory, with a primary research focus on communication network agents.
Responsibilities: Focused on engineering applications of LLM technologies in communication networks, including configuration translation,

2025/11 - 2026/01	Guangzhou Zhiyong Artificial Intelligence Research Institute	Position: LLM Application Engineer
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Responsibilities: Provided technical solutions for agent projects, developed Python proof-of-concept demos to validate feasibility, and built production-oriented agent applications. During the internship, I independently completed multiple projects, including a VL-model-based table content extraction agent, an image translation agent translating any language into Chinese, and TTS/STT agents for Indian English and Chinese.

Organization: A China subsidiary of the France-based Thales Group, responsible for intelligent card operations. Responsibilities: Developed robotic process automation (RPA) workflows. I coordinated cross-departmental requirements, designed four RPA programs for the company, and modified and optimized database business logic for tasks such as automatically searching data in Oracle systems and filling in and downloading Work Orders and BOM Routing. These programs reduced repetitive manual work, lowered the risk of human error, and improved production management efficiency.

Professional Knowledge: Solid foundation in mathematics (optimization theory, calculus, algebra), statistics (probability theory and mathematical statistics, regression analysis, statistical inference), and computer science (data structures and algorithms, computer networks, microcomputer principles).

Programming Languages: Strong Python skills; proficient in LangChain, PyTorch, NumPy, and related libraries; experience in model training, fine-tuning, and agent development; familiar with database systems and SQL-based database management; familiar with C++.

English: IELTS Academic: 7.0; CET-6: 582.

2024 MCM (Mathematical Contest in Modeling): Finalist (global top 2%)

2025 Outstanding Undergraduate Thesis, South China University of Technology (university-wide top 1 %)

2025 First Prize, SCUT "Meet Xiaomian Innovation Scholarship" for the Student Scientific and Technological Innovation Competition (university-wide top 1%)

2023 First Prize (1st place), Guangzhou University Mathematical Modeling League

Additional awards in mathematical modeling, scholarships, outstanding student leader honors, etc.